and 7, General Letter A, Appendix D) I directed that limits were to be fixed for each town, cantonments, and Civil lines, which should include all suburbs and areas in or upon which people found on the night of the Census would properly be included in the population of the town. I pointed out that in the case of municipalities, municipal limits would ordinarily suffice; while in that of cantonments, cantonment limits should be followed. I directed that where the town had arable land attached to it and constituted a revenue-paying estate, the area included within the urban limits just described should be treated throughout as quite separate from the remaining village area, and from any hamlets that the latter might contain. Subsequently, at Mr. Elliott's desire, I stated that the urban limits of municipalities should correspond with municipal limits except under very special circumstances.

These instructions were, I am glad to say, not carried out; at least not in the manner intended by me. I had meant that limits were to be fixed for the urban area of the town of Dehli, for instance; and that everything within that area was to be labelled "Town of Delhi" without further specification. But this would have deprived us of all details of population for suburbs; and thus, while we should have been saved an immensity of trouble in compilation, we should have lost much very valuable information. It may be, even now, that the limits of certain towns and the population returned for them may include suburbs for which it would have been well to have had separate figures. But a glance at the Appendix to Table XIX for British Territory will show that such a case, if it occurs at all, is at least the exception.

846. The way in which I would treat the matter on a future occasion is this. The external urban limits in each case would be governed by the principles to be presently laid down in the section on Tabulation. Within those limits I would fix the boundaries of (1) the town proper, (2) each suburb or group of suburbs for which it was desirable to have separate figures, separating as a general rule every suburb that bore a distinct name, and which it was at all possible to separate. The town proper might be taken by wards, if such divisions existed. The limits of each section should include all encamping grounds, waste land, or other urban spaces which properly attached to it, so that the area of all the sections taken together should include all that would be included within the external urban limits.

I would then make my Census blocks follow these sections, so that no block should as a rule contain any part of more than one section. If a block contained two small sections I would have a separate book of schedules for each, and cases where boundaries crossed would be similarly provided for. For instance, part of the Ambála Civil lines lies within and part without municipal limits; and, as I have not separate figures for the two parts, I have had to treat the population of the whole as part of the population of the municipality in the present Census. In this particular case the error is absolutely insignificant; but it and similar cases would be met by having separate books of schedules for the intra and extra-municipal portions. The arrangement above described would give us separate figures for every separate group of houses which could possibly be considered as falling within the limits of the town.

The question which of these groups should be reckoned to form part of the town could then be dealt with at leisure. I shall discuss the principles on which I think it should be decided when treating of the classification of population as urban and rural in the section on the Tabulation of the figures.

847. Size of Census sub-divisions.—At the Panjáb Census of 1868 each block included 100 families (then called houses), and each circle 20 blocks in towns and 10 to 15 in scattered village areas. The Committee of 1877-78 recommended that each block should not contain more than 60 families, or 300 souls; and that a circle should not contain more than 10 blocks, supervising agency being still further strengthened on the night of the Census. I pointed out that the agency available and the distribution of the population would vary so greatly that no hard-and-fast rule could be laid down; but that the "limits "for a block prescribed at the last Census should not ordinarily be exceeded; while the supervisors should be "appointed as nearly as possible on the scale recommended by the Committee." I also recommended that not more than four circles should be placed under a single superintendent. Abstract No. 143 on the opposite page shows the average number of people included in each town and village block respectively, and the average area of the latter in each district in the Panjáb. Column 9 shows the area per block after deducting the waste lands belonging to Government, which are not indeed wholly uninhabited, but of which the nomad pastoral population is exceedingly scanty, and required special arrangements for their Census.

I had hoped that district officers would avail themselves fully of the license I gave them in my circular, and would work up to the full limits of 500 souls per block. But I find that they almost universally speak of 300 souls as "the prescribed limit," while the table shows that they did not take me at my word; and in one district I found small hamlets of only 8 or 10 houses made into separate blocks, and villages of less than 400 inhabitants divided between two enumerators. This I had altered at once; and in fact the figures of the Abstract, representing as they do the final number of blocks, do not adequately represent the degree of sub-division which was at first attempted; for blocks which had at first been separated were in many cases thrown together before the Census, owing either to my representations or to the impossibility of finding enumerators for them. Of course in hill districts like Simla, Kángra, Hazára, and Kohát, where the population is distributed over small hamlets lying at considerable distances from each other, and where travelling is exceedingly difficult, it is unavoidable that each block should include only a small number of people. And generally, distance is a very important factor. But there seems no sufficient reason why the average population of a village block should have been below 300 in Hushyárpur and Muzaffargarh, and below 350 in Dehli and Gurdáspur; or that of a town block below 300 in Ludhiánah, Hushyárpur, Gurdáspur, Multán and Dera Ismáil Khán, and below 350 in Dehli, Jálandhar, Siálkot, Gújránwála Ráwalpindi, Gújrát, Sháhpur, Montgomery, Muzaffargarh, Derah Gházi Khán, Pesháwar, and Hazára.

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Part I.—Preliminary Operations.

Abstract No. 143, *showing the* Number and Size of Census Blocks *for* Districts.

1	2	3	4	5	6	7	8	9	10
District.	Number of Blocks.			Average number of persons per block.			Average area of village blocks in acres.		Index to
	Villages.	Towns.	Total.	Villages.	Towns.	Total.	Total area.	Corrected area.	
Delhi	1,288	653	1,941	341	312	332	634	634	Delhi
Gurgaon	1,307	115	1,422	436	628	451	949	949	Gurgaon
Karnal	1,052	180	1,232	517	435	505	1,457	1,457	Karnal
Hissar	1,090	181	1,271	408	421	410	2,158	2,158	Hissar
Rohtak	1,134	250	1,384	400	398	401	1,022	1,022	Rohtak
Sirsa	713	75	788	312	413	321	2,697	2,697	Sirsa
Ambala	2,040	319	2,359	454	440	452	806	806	Ambala
Ludhiana	1,532	295	1,827	350	282	339	574	574	Ludhiana
Simla	84	118	202	250	187	2 3	137	137	Simla
Jalandhar	1,767	414	2,181	369	331	362	479	479	Jalandhar
Hushyarpur	3,268	285	3,553	253	260	254	427	427	Hoshyarpur
Kangra	3,098	109	3,207	228	225	228	1,874	1,874	Kangra
Amritsar	1,602	441	2,043	440	426	437	629	629	Amritsar
Gurdaspur	2,260	344	2,604	328	237	316	516	516	Gurdaspur
Sialkot	2,171	247	2,418	428	340	419	577	577	Sialkot
Lahore	1,331	443	1,774	541	462	521	1,754	1,547	Lahore
Gujranwala	1,655	219	1,874	321	370	329	1,000	696	Gujranwala
Firozpur	1,454	154	1,608	402	429	405	1,221	1,221	Firozpur
Rawalpindi	2,816	284	3,100	261	300	265	1,005	858	Rawalpindi
Jahlam	1,528	135	1,663	346	453	354	1,628	1,421	Jahlam
Gujrat	1,664	130	1,793	389	326	384	759	759	Gujrat
Shahpur	1,023	163	1,186	362	317	355	2,433	2,095	Shahpur
Multan	1,360	297	1,657	341	297	333	2,767	1,352	Multan
Jhang	880	84	964	407	440	410	4,047	1,352	Jhang
Montgomery	937	69	1,006	430	312	424	3,805	1,352	Montgomery
Muzaffargarh	1,430	63	1,493	221	347	227	1,405	1,188	Muzaffargarh
Derah I. Khan	1,344	209	1,553	293	229	284	4,422	3,855	Derah I. Khan
Derah G. Khan	709	135	844	452	320	431	4,057	3,057	Derah G. Khan
Bannu	589	55	644	520	469	516	4,203	4,053	Bannu
Peshawar	1,130	391	1,521	411	329	399	1,418	1,418	Peshawar
Hazara	1,301	62	1,423	285	303	286	1,429	1,315	Hazara
Kohat	452	35	487	361	519	373	4,008	3,911	Kohat
Total	46,089	6,954	52,983	357	350	356	1,487	1,256	

848. Taking the Province as a whole, I think that the blocks were too small; and more especially in towns, where the average number of souls per block is even smaller than in villages. I attribute the fact partly to a common disinclination, which I found to exist in many districts, to include two villages in one and the same block. It is of course absolutely necessary that the block should be of such a size, both as regards the number of people included within it and the distance between the points to be visited, that a single man can easily visit and correct the entries for every house in it on the night of the Census. But so long as this precaution is duly observed, the larger the blocks the better. An excessive number of blocks increases the difficulty of finding enumerators, compels recourse to uneducated agency writing strange and illegible characters, renders supervision more difficult and uniformity less easy of attainment, increases the labour and expense of compilation, and obliges the Government to call upon a larger proportion than necessary of the non-official population to perform a very irksome duty. In many districts the very greatest difficulty was experienced in finding enumerators for the outlying tracts; and in not a few cases men unacquainted with the locality and its inhabitants had to be sent out from the towns, and their expenses paid from district funds. Yet in those very districts I not unfrequently found that two blocks had been made where one enumerator would have been amply sufficient. In the towns themselves qualified agency existed in plenty; but here the houses lie so close together that it would have been well, by increasing the size of the blocks, to reduce as much as possible the demand upon private individuals for assistance in the operations. I found that, in the City of Lahore, the enumerators in the portions which I visited had generally finished their work by 10° 30′ or 11° on the night of the Census; and I believe that where the houses lie in a compact mass, a single enumerator can easily manage 500 to 600 people, for he has not to take their Census *ab initio*, but merely to check and correct a Census already taken at the preliminary enumeration. Where the houses lie far apart, as in the case of scattered hamlets and villages, the distance to be travelled, and not the number of houses to be visited, is the most important fact. Mr. Stogdon of Lahore, Major Plowden of Kohát, and Colonel Parsons of Jahlam would increase the size of the block, Mr. Stogdon remarking that "it is difficult to get enumerators, and it is better to have intelligent enumerators and large blocks than enumerators of no capacity and small blocks." Major Parker of Hissár, Mr. Hawkins of Amritsar, and Mr. Fitzelle of Sháhpur are of opinion that a block should include at least 100 to 125 families "which a single man could easily visit and correct the entries of; while where the block lies wholly in one town or village it might safely include 125 houses irrespective of families." Mr Benton of Karnál writes:—

"A large number of the enumerators, such as the patwáris lately employed in the settlement and many others, were admirably "qualified for the work, and although to meet the attainments of men generally available, it is no doubt desirable to keep the

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Part I.—Preliminary Operations.

"average size of blocks low, I do not hesitate to say that such men could easily perform the duties satisfactorily on the night of "the final enumeration for 200 or 300 houses, provided they were all together.

"I would retain the average size of block at 60 houses or 300 persons as suited to enumerators of the meanest capacity, but I "would make it optional to make over two or three of such blocks to properly qualified persons, because so long as preliminary enu- "meration is had recourse to, an active and intelligent man could easily attest 1,000 entries, when the houses are contiguous, in three "or four hours. The work of enumeration was almost completed in Pánipat before moonrise, that is to say before 8° 20' P.M."

Mr. Gardiner of Gújránwála indeed would decrease the size of the block if he could get enumerators enough; but he gives no reason for his opinion.

The statement of Census agency given on the opposite page shows that the proportion of supervisors to enumerators was as a rule largely in excess of that suggested in my instructions, in many cases being more than twice as great, and only in Lahore, Montgomery, and the villages of Derah Ismáil Khán falling materially below it. The number of blocks placed under each supervisor was on an average—

In the villages	8⅘
In the towns	7⅓
In the Province	8⅘

The average number of circles allotted to each superintendent was—

In the villages	3¾
In the towns	3
In the Province	3⅔

Here again, the agency was most numerous in the towns; and this was I think right; for the town enumerators, while or rather because more highly educated, are less intimately acquainted with the people they have to enumerate, more satisfied with their own ignorance, and less inclined to follow instructions, while the population they have to deal with is far more varied and complex. Thus supervision, while easier to exercise, is far more necessary in the towns than in the villages.

849. Statistics of Census agency.—Abstract No. 144 on the opposite page shows the number of the Census staff, and the sources whence they were drawn.

The results may be abstracted as follows :—

		Villages.	Towns.	Total.
Number of	Superintendents	1,444	314	1,758
	Supervisors	5,440	959	6,399
	Enumerators	46,029	6,954	52,983
	TOTAL	52,913	8,227	61,140
Official Staff	Superintendents and Supervisors	2,419	750	3,169
	Enumerators	3,670	1,748	5,418
	TOTAL OFFICIAL	6,089	2,498	8,587
Village accountants.	Supervisors	3,067	81	3,148
	Enumerators	4,223	100	4,323
Village headmen	Enumerators	2,759	59	2,818
	TOTAL SEMI-OFFICIAL	10,049	240	10,289
Shop-keepers	Enumerators	10,958	342	11,300
Schoolboys	Enumerators	3,084	859	3,943
Other non-officials.	Superintendents and Supervisors	1,398	442	1,840
	Enumerators	21,335	3,840	25,181
	TOTAL NON-OFFICIAL	36,775	5,489	42,264

To furnish so large an agency, every class of the community which had any pretensions to education or intelligence was called upon to contribute. Almost every official in the district was employed in Census duties, the supervising staff being chiefly drawn from this class; while private individuals formed the great mass of the enumerating agency. The superintendents were for the most part Munsifs, Náib Tahsíldárs, Qánúngos and their Assistants, Deputy Inspectors of Police, superior office staff, Chief *Muharars* and head-masters of schools, Assistant Surgeons, *Daroghas*, members of committees, Government pensioners, Honorary Magistrates, merchants, *Zaildárs*, *Jágírdárs*, and rural notables and men of influence. The supervisors were drawn from the classes mentioned above, supplemented by officials from the district revenue, judicial, and municipal establishments, sergeants of police, district school and post masters, native doctors, village accountants and headmen, *jágírdár's factors*, intelligent pupils from the higher schools, respectable traders, and land-owners and others of intelligence. The class of enumerators comprised every clerk or similar official who could be spared and whose services had not already been utilised in the higher grades; and to these were added a vast body of non-official agency, stamp-vendors, petition-writers, candidates for employment, pensioners of sorts, retired *Patwáris*, merchants' clerks, *Jágírdárs'* servants, *dharwáis*, senior school-boys and monitors, *Pandits*, *Maulvis*, village *Mullás* and *Qázis*, shop-keepers, village headmen, and land-owners who could read and write. The services of even illiterate but intelligent *zamindárs* were often utilised by attaching to them school-boys as amanuenses.

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Abstract No. 144, *showing* Census Agency *for* Districts.

Serial No.	Name of Division	No. of Supervisors.			No. of Enumerators.								TOTAL VILLAGES AND TOWNS.												Name of District.	Serial No.
		Officials.	Non-officials.	Total.	Village Accountants.	Village Headmen.	Other officials.	Village Shopkeepers.	School-boys.	Other non-officials.	Total.	Officials.	Non-officials.	Total.	Village Accountants.	Other officials.	Non-officials.	Total.	Village Accountants.	Village headmen.	Other officials.	Village Shopkeepers.	School boys.	Other non-officials.	Total.		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	1
1	Delhi	36	14	70	200	44	32	285	110	33	138	518	104	376	1,288	60	32	92	111	65	187	519	222	867	1,941	Delhi
2	Gurgaon	30	7	37	59	49	38	146	245	68	153	273	137	419	1,307	34	7	41	247	55	188	278	146	490	1,423	Gurgaon
3	Karnal	33	4	37	55	38	14	107	171	36	141	99	49	339	1,052	28	6	34	175	43	128	99	61	659	1,232	Karnal
4	Hisar	35	5	30	70	21	22	113	141	36	74	396	36	397	1,050	38	8	46	145	37	134	403	79	427	1,231	Hisar
5	Rohtak	45	8	53	148	28	8	184	275	72	118	425	84	424	1,134	34	8	42	84	33	130	454	133	505	1,384	Rohtak
6	Sirsa	18	...	18	16	22	4	112	108	11	307	153	18	84	713	23	...	23	108	46	348	154	27	105	788	Sirsa
7	Ambala	...	...	...	24	165	57	246	193	61	170	109	541	905	2,040	...	1	64	198	63	218	135	586	1,164	2,359	Ambala
8	Ludhiana	48	...	48	76	99	6	181	254	68	212	448	242	368	1,533	63	1	64	254	68	200	448	350	447	1,817	Ludhiana
9	Simla	2	...	2	4	6	2	13	1	10	10	...	10	53	84	7	...	7	1	10	40	...	19	132	202	Simla
10	Jalandhar	61	6	67	168	28	18	214	175	114	103	149	103	1,033	1,767	78	10	88	178	118	163	180	383	1,311	3,181	Jalandhar
11	Hoshyarpur	76	25	101	233	132	54	419	141	181	387	154	342	2,063	3,268	85	27	112	238	184	404	388	388	2,202	3,583	Hoshyarpur
12	Kangra	65	7	72	158	98	61	317	11	369	63	1,384	40	1,331	3,098	71	8	79	159	104	184	1,403	49	1,286	3,207	Kangra
13	Amritsar	32	51	83	153	53	27	210	271	51	109	141	79	951	1,602	44	56	100	153	51	157	141	100	1,333	2,043	Amritsar
14	Gurdaspur	33	57	80	234	26	6	298	145	182	133	90	49	765	2,266	34	66	100	238	185	238	904	75	966	2,664	Gurdaspur
15	Sialkot	33	56	88	224	33	48	278	142	137	91	958	54	789	2,171	43	62	104	239	153	153	982	64	939	3,418	Sialkot
16	Lahore	24	4	28	38	48	27	113	220	57	124	238	93	604	1,331	32	22	54	43	85	226	296	113	854	1,774	Lahore
17	Gujranwala	33	5	38	52	70	49	171	198	47	71	283	51	1,005	1,655	39	10	49	54	51	119	334	75	1,004	1,874	Gujranwala
18	Firozpur	42	12	54	54	98	33	185	227	33	51	323	55	700	1,434	45	12	57	136	51	120	334	80	812	1,668	Firozpur
19	Rawalpindi	64	28	92	160	66	77	303	134	332	99	964	56	1,240	2,816	75	30	105	161	145	333	1,600	86	1,390	3,100	Rawalpindi
20	Jhelum	49	13	62	182	13	10	205	37	110	72	115	100	1,091	1,528	54	14	68	187	20	113	196	114	1,198	1,691	Jhelum
21	Gujrat	32	11	43	117	14	14	166	75	109	99	645	101	635	1,664	37	11	48	120	21	110	196	113	1,199	1,794	Gujrat
22	Shahpur	28	3	31	98	17	9	124	38	69	81	270	53	506	1,023	31	4	38	104	25	70	299	73	586	1,186	Shahpur
23	Multan	33	8	41	71	24	47	142	171	83	95	438	58	515	1,360	38	14	52	72	45	83	447	133	643	1,657	Multan
24	Jhang	10	7	17	26	37	30	93	133	27	53	30	67	570	880	24	7	31	18	43	29	30	85	394	964	Jhang
25	Montgomery	35	7	42	135	47	32	81	138	38	55	90	80	539	937	36	11	47	5	53	28	91	83	374	1,096	Montgomery
26	Muzaffargarh	38	7	65	135	19	14	168	45	83	115	112	73	1,062	1,430	62	8	70	136	83	178	83	95	1,035	1,493	Muzaffargarh
27	Dera I. Khan	28	2	30	72	38	14	124	122	87	84	221	153	674	1,344	37	6	43	73	59	172	324	176	767	1,553	Dera I. Khan
28	Dera G. Khan	11	11	22	11	23	44	78	117	62	83	40	12	395	700	22	11	33	120	41	150	40	14	453	844	Dera G. Khan
29	Hamun	5	...	5	14	38	14	66	132	22	57	43	25	310	589	6	...	6	136	42	89	44	28	343	644	Hamun
30	Peshawar	27	4	31	74	17	38	129	217	103	61	686	20	143	1,130	33	8	41	216	37	173	596	32	387	1,521	Peshawar
31	Hazara	27	27	54	114	1	46	161	107	71	40	393	103	852	1,301	31	26	57	3	46	59	292	113	884	1,423	Hazara
32	Kohat	10	...	10	22	21	...	43	28	9	220	72	18	111	452	11	...	11	28	...	233	72	33	112	487	Kohat
	Total	1,067	387	1,444	3,067	1,362	1,017	5,440	4,223	2,759	3,670	10,958	3,084	21,335	46,020	1,875	483	1,758	3,148	1,894	1,357	6,390	4,393	2,818	5,418	11,300	3,943	25,181	54,983	

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Part I.—Preliminary Operations.

850. Aid afforded by independent Departments.—At the very commencement of operations the Panjáb Government issued a circular to all Heads of Departments[1] pointing out that "every available "officer in every department of Government who can be relied on for efficiently conducting supervision "will be expected to render assistance; and Heads of Departments are requested to place the services of "their staff unreservedly at the disposal of the District Officers for this purpose, so far as may be compatible "with the conduct of necessary business." The manner in which these orders were carried out varied much in various Departments and parts of the Province. In those districts where a settlement was in progress, *viz.*, Sirsa, Lúdhiána, Jálandhar, Hushyárpur, and Ráwalpindi, the settlement establishment afforded a ready-made supervising agency of the utmost value. The Deputy Commissioners and Settlement Officers came to an agreement about the division of the work in each case; and the most practical plan was that by which the latter undertook, in communication with the former, the whole of the purely village Census, leaving the remainder to the district staff proper. The Forest Officers rendered most valuable assistance in the difficult mountain tracts in which their duties lie; and the Customs officials in Gurgáon worked most zealously for us. The staff of the Western Jamna Canal was of great use in the final Census of some districts; while the postal, educational, and medical departments as a rule responded cordially to the call made upon them. But complaints of want of willingness to help in the work are not altogether absent. The Panjáb Northern State Railway refused to assist; in Gurdáspur the Canal staff formed a solitary exception to the willingness of other Departments; in Gújrát the educational officers were obstructive; while in Jahlam little aid could be obtained outside the district staff. In Pesháwar, the staff drawn from other Departments "did little, and never made any attempt to learn the working of the Census machinery." Major Nisbet writes from Simla:—

"One of the weakest points in the general arrangements was, the little assistance that could be obtained from other Depart-"ments for Census work, notwithstanding the orders Government promised to issue on the subject. The usual excuse was, that "office hands were generally fully employed in their own special duties. On a future occasion it might be well for Government to "ascertain beforehand the assistance obtainable from different offices, and determine the sufficiency of excuse offered for neglecting "to afford such aid."

I think it would have been well had Government issued a second circular early in December, again insisting upon the importance of help being cordially rendered. And much can be done by the Deputy Commissioner to make matters easy for other Departments. Mr. Benton of Karnál for instance used the Canal staff as enumerators only. He required them to attend at their blocks from 26th December to 2nd of January and again from 15th to 18th February, and he had the supervisors ready on the spot with the papers to explain their duties to them. In any case the brunt of the work must fall upon the Deputy Commissioner and his subordinates. I called for returns from the districts showing the amount of assistance rendered by other Departments; but unfortunately the figures are rendered meaningless by the fact that most of the Deputy Commissioners have included under that heading the district staff in the Police, Educational, Medical, Jail, Registration, and similar departments, which, being directly subordinate to the Deputy Commissioner, should have been counted as District Staff. The Government of India forbade us to employ the rank and file of the police on Census duty; and though the exception deprived us of the services of a large staff distributed all over the country, yet I think it was absolutely necessary.

851. Preparation of the preliminary record by village accountants.—It will be noticed that of the village accountants, nearly half were utilised in supervision. As a fact the great mass of the preliminary records in the villages was I believe made by these men, whether as supervisors or as enumerators. At an early stage of the operations I pointed out[2] that two classes of agency would be employed for the work of filling up the schedules, as it would sometimes be impossible to obtain for the preliminary record the services of the man who was to conduct the final enumeration on the Census night, and the staff which prepared the former would have to be supplemented for the latter operation. I had wished to put the point more strongly, and to say that in villages this would very generally be the case: but the Panjáb Government was of opinion that the objections to having the preliminary record prepared by one man and the final Census taken by another were so great that, though such a course would doubtless be unavoidable in some cases, yet it was well to treat it as a quite exceptional and deplorable necessity.

In the towns it is not only necessary but easy to arrange that the same man shall perform both operations. But in the villages this is not the case; and everything I have seen and heard in connection with this Census has convinced me that it is better to prepare the preliminary record by the best available agency, and where necessary to this end, to sacrifice the advantage of having the same man make and finally correct the entries. Of course it is most advisable that, where a really competent agent can be secured for the final enumeration, he shall also fill up his schedules in the first instance, as he thus obtains an intimate knowledge of his block and its inhabitants. But in the villages it is too often simply impossible to obtain really good men in sufficient numbers to take the final Census. The fact that of 40,000 village enumerators, 11,000 were village banyas, 3,000 village headmen, 3,000 school-boys, and 21,000 non-officials resident (for the most part, at any rate) in the villages, speaks for itself. For the preliminary record on the other hand, which is made more or less at leisure, it is comparatively easy to obtain a sufficient and competent staff: in fact the village accountants might almost prepare it unaided. Now the preliminary record is, in the villages at least, the very backbone of the Census; in fact it might almost be said to be the Census itself. I do not believe that the alterations made on the night of the Census affected 5 per cent. of the village population. But supposing the proportion to have been double this, and that in a village of 500 souls, the enumerator cut out 25 as having died or gone away, and entered 25 new men as having been born or come to the place since the preliminary record, I would prefer to have the 500 entries correctly made in the first instance, and to take the chance of the enumerator making a mess of his 25 new entries—the 25 excisions he could not well go wrong with—rather than have

[1] Section 3 of Secretary to Government's No. 7, dated 27th August 1880, Appendix D.
[2] Section 11, General Letter A, Appendix D.

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Part II.—The Enumeration of the People.

end of each family would have wasted much room and served no useful purpose. On this subject Mr. Wilson of Sirsa writes:—

"I think that too much importance is given to the 'house,' which is often an accidental conglomeration of families, and too "little to the 'family' which is bound together by natural ties of great force. For instance if a baby was born after the preliminary Census and before the 17th February, it was then entered at the end of the house, perhaps a long way from the entries "relating to its father and mother and below entries relating to a totally distinct family. I would class individuals into families, "and for this district at least would not attempt to class them into houses."

I agree with him as to the relative statistical importance of houses and families. But the record of families is taken from the entries of the preliminary record; and the addition of the baby at the end of the house did not affect the figures. Moreover, I propose (in section 844) to take these figures in future from the house-lists, and not from the schedules. If however it is desired to take statistics for families from the schedules, it would be well to direct the enumerator, in cases where whole families had come to the house since the preliminary record, and especially in hostelries and the like, to separate off such families from isolated individuals in the new entries made on the Census night by two lines across the name column, just as families are divided off at the preliminary record. This would make the record of families more accurate; but the primary object of distinguishing families in the schedules is, not to obtain statistics, but simply to facilitate checking.

All excisions and entries made on the night of the Census were made in red ink. Mr. Elliott feared that some difficulty would be experienced in supplying this to the enumerators; but this was not found to be the case. The common magenta aniline dye (*shíshah ká rang*) is cheap, sufficiently permanent for the purpose, goes a long way, and is procurable in the bazaar of any considerable town. One Deputy Commissioner estimated for some hundreds of rupees worth, enough to dye all his enumerators scarlet; but this was unnecessary. The colour was distributed through Tahsíldárs to the supervisors and by them to the enumerators. And the use of red ink on the Census night was exceedingly valuable, as it enabled the supervisors to distinguish at once the entries of the preliminary record, which he had already tested, from those of the final Census which he had still to test, and prevented enumerators from playing tricks with the entries.

865. It was proposed, at a late stage in the operations, to instruct the enumerators to distinguish residents from visitors in the schedules. The late date of the proposal would of itself have prevented its adoption in the Panjáb. But in any case I do not think it a good one. The permanent population of any town or village=its population on any night+absentees on that night—visitors on that night. Now, though it would be easy enough to distinguish the visitors, yet it would be practically impossible to obtain any record of the number of absentees under any system compatible with a synchronous Census, in which each person is enumerated wherever he may happen to be on a given night. In large towns and even in villages, save under special circumstances such as the occurrence of a fair, the visitors and absentees would be about equal in number at any given time; and the numbers actually present on the Census night would probably give a far more accurate idea of the normal population of the town than would those numbers after the exclusion of visitors. If it were really desired to obtain an enumeration of the permanently resident population of each place, the proper plan would I think be, to accept my proposal that temporary visitors should not be entered at the preliminary record (see section 862), and further to direct that all persons only temporarily absent should be entered as if at home in the preliminary record. These last would of course be cut out again at the final Census if they had not returned meanwhile; but if the number of excisions and additions made in red ink on the Census night were tabulated, they would give by a very simple sum the permanent population as it stood at the time of the preliminary record, births and deaths occurring between that time and the final Census not being taken into account. The only danger would be that the names of persons temporarily absent at the preliminary enumeration might be left standing at the final Census, although they had not returned meanwhile.

On the night of the Census the supervising staff were strengthened wherever possible (chiefly, I suspect, in the towns) and each enumerator was visited while at work wherever distance would allow of its being done. Where this was impossible, the supervisors were instructed to direct their attention to the larger villages with miscellaneous populations, and to those on frequented roads where many travellers might be expected to stop. Next morning they started off to go round their circles and test on the spot every excision or entry that had been made on the Census night. They were allowed to cut out or add any person who appeared from enquiry to have been wrongly left in or omitted; but they were not to alter the details recorded concerning any person rightly entered, and who was not before them, unless they could make perfectly sure of what should have been recorded.

This completed the operations, so far as they concerned the enumeration of the people.

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PART III.—SPECIAL MEASURES.

866. Municipalities.—The Census of the Municipalities, which embrace almost all the towns with more than 5,000 souls, and many with smaller populations, was taken through the Municipal Committees. In the large Municipalities the arrangements were most thorough and complete; and in several cases advantage was taken of the opportunity to prepare permanent maps and house-registers. The cost of the Census up to the time that the books were sent to the tabulating offices was borne by the municipal funds in all Municipalities of the first and second classes, and in such of those of the third class as had at the Municipal Census of 1875 a population of not less than 10,000 souls; forms and instructions, however, being supplied at Government expense. But even here, where funds were available, it was not found necessary to employ paid agency save for the preparation of house-lists in the city of Dehli, and for the whole operations in the city of Amritsar. Agency was, of course, easy to procure; and the only respect in which the Census of the large cities was not specially accurate was the entry of religion, caste, and clan. Here the educated city enumerators knew far too much to record accurately; and from what I saw and heard, I suspect that the poorer classes in cities do often lose in a far greater degree than do those in the villages, their regard for and memory of caste distinctions. A not inconsiderable proportion of them often consists of a miscellaneous collection of low-caste men from the North-West Provinces who are indifferently classed as Púrbias by the people of the Panjáb; and they themselves seem in many cases to have grown to consider the classification as sufficient.

867. Cantonments.—The Census of cantonments was taken through the Cantonment Magistrate; and the cost was borne, as in Municipalities, by cantonment funds. The degree of success with which the work was done varied immensely, and depended wholly upon the character of the Cantonment Magistrate. Some of the very best work I saw was in cantonments. In other cases the Magistrate was lazy and indifferent, the greatest pressure had to be put upon him to induce him to move at all, and in one cantonment at least the Census would hardly have been taken at all if the district officers had not stepped in at the last moment and done the work themselves. In others again no care was taken in selecting the agency, no supervision of any sort was exercised, and the staff had to be threatened with criminal proceedings before anything could be done. The position is a difficult one. The Cantonment Magistrate is subordinate to the district officer only in his judicial and not in his executive capacity, while the Cantonment Committee which control his action in the latter capacity is of course wholly independent. But, as the appointment of the Magistrate rests with the Panjáb Government, I think on a future occasion a lazy and indifferent man might be given clearly to understand that the work had got to be done, and that he must do it, and do it properly. In Jahlam, Talágang, and Naushahra the whole work, excluding of course the Census of military limits, was done by the district staff with the consent of the cantonment authorities, who placed the cantonment staff at their disposal. Where this arrangement is agreed to nothing could be better. Messrs. Beckett and Lawrence of Peshâwar both recommend that either the work should be made over absolutely to the civil authorities, or that they should be relieved of all responsibility for it. Probably the best way would be to empower the Deputy Commissioner, in case proper arrangements are not made by the cantonment authorities, to do the work himself by paid agency and charge the cost to cantonment funds. The population of cantonment bazaars is more mixed than any other population in the Province, and its Census presents very peculiar difficulties as to the entries in some of the columns. It is, however, insignificant in extent.

868. Military Census.—It was, of course, absolutely necessary to take the Census of troops through their own officers. At the same time it was advisable to throw as small a burden as possible upon the military authorities. It was therefore decided that the Census of only "regimental lines, quarters, barracks, or other premises in which troops or others subject to military discipline are quartered" should be taken by the military authorities; "that of regimental and sadr bazaars, and of all parts of cantonments except "the quarters just mentioned, being taken by the civil authorities acting through the Cantonment Magis-"trate." Civil officers were directed to fix, in consultation with the military authorities, the limits within which each was respectively responsible for the Census. A proposal was made to define the several responsibilities of the two departments by making the military authorities answerable for the Census of certain specified *classes of persons*, and of them alone. And in the Railway Census I found a similar standard proposed in one case, whereby to distribute the responsibility for the Census among the several departments into which the railway staff was divided. I am convinced that any such rule must result in confusion and error. Where two persons have to take a Census side by side, the only safe plan is to fix *topographical boundaries*, and to make each responsible for the enumeration of all persons of whatever description found within his limits. At the same time, to reduce the risk of error, the military authorities were requested to clear all military limits, between sunset and sunrise of the Census night, of all persons not temporarily or permanently living within them; and this, I think, was useful.

869. The general plan of preliminary operations was somewhat simplified for the military Census. The rules on the subject will be found in Appendix D. Military quarters had been treated as houses in the general Census of cantonments, and entered as such in the house-lists [1]. This, as Mr. Elliott pointed out, was a mistake; and it would have been better to have treated them as blocks. It had not been decided, when I issued the instructions, within what limits the military Census should be confined; but in future it might be explained to the civil authorities that the military Census would be confined within the narrow limits already described, and that such limits should be treated as blocks for the Census of which the military authorities would be responsible. No separate index-register was prepared by the Military Department;

[1] Section 9, General Letter A, and Rule 6, Part I, Instructions to Supervisors, Appendix D.

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Part III.—Special Measures.

no maps were needed; and a simplified form of house-list was used. But of course the schedule and the instructions for filling it up were the same as those used elsewhere, and a preliminary record was made. As soon as the entries of the Census night had been finally examined and corrected, the schedules were made over to the district office, and the military authorities had no further concern with them. Troops on the march were censused by their officers, the schedules being sent, in the case of a detachment to the regimental headquarters, and in the case of a regiment to the station at which they were last cantoned. The schedules of a regiment on its return from service beyond the frontier, which was censused at Karnál, were included in the population in that place. This was done by order. I should have preferred to include them in the population of the nearest cantonment.

A European civil officer was appointed in each case to advise the military authorities, and to exercise such general supervision over their operations as was necessary to secure uniformity in the results. It was found exceedingly difficult to get the authorities to move in time. It was not that they were indifferent or neglectful; in fact as far as I can judge (and I have examined all the English schedules page by page) the Census of military lines was carefully taken; far more so than that of many cantonments. But the military Census was such a simple matter that the whole of the work might have been done within the week immediately preceding the Census night. Thus the preliminary arrangements were put off to the last. But printed forms and instructions had to be provided; and in some cases the indents for these had not been received by the beginning of February, and I had to estimate the probable requirements as well as I could, and send off such a supply as I thought would certainly be sufficient. I at least, if no one else, should have been saved much anxiety if the operations had not been so delayed, that not till the day before the Census did I receive news of the completion of the preliminary record for the Province.

870. The Khaibar troops.—At the last moment Mr. Elliott informed me that he had, in consultation with the Government of India, decided to include the troops and camp-followers then lying in the Khaibar Pass in the Census of the Indian Empire; as, though situated beyond British Territory, they formed an integral part of its population. I accordingly sent them schedules and instructions. When the schedules were returned to the tabulating office it was found that for the most part only four columns, namely those showing name, sex, religion, and occupation, had been filled in. The military authorities informed the Deputy Commissioner that they had authority for thus curtailing the entries; but, though I reported the fact, I am unaware what orders had been issued on the subject. In my tables the Khaibar has been treated as an eleventh division, and its people included in the population of the British Panjáb whence they were for the most part drawn. Where the entries in the schedules were left blank, the numbers are shown as "unspecified."

871. Public institutions and gangs of workmen.—The Census of all public institutions, such as gaols, police-lines, dispensaries, hospitals, asylums, and the like, and of gangs of workmen employed on railways, canals, &c., was taken through the officers in charge. They were treated as single houses in the house-lists [1]. Mr. Elliott was of opinion that they would better have been treated as blocks; but I think that except where very large, they were best treated as houses of which the occupier filled up his own private schedule. Schools and other similar institutions, and police, canal, forest, customs, and other outposts were treated in a similar manner. Gaols [2], outposts, and similar buildings were entered as houses in the general house-list of the blocks in which they were situated, and their resident families recorded therein. Separate house-lists were prepared for gangs of workmen on large works. In both cases the preliminary record was deferred to within a few days of the Census on account of the unstable nature of the population.

872. Private schedules.—The Government of India directed that private schedules, to be filled up by the householder himself and not by the enumerator, were to be given only to "highly-educated people and members of the European community," as it had been found at previous Censuses that they were often filled in most correctly. They were also distributed to all schools, outposts, and similar houses mentioned in the preceding paragraph. They had instructions for filling them up printed on the back, and a specimen will be found in Appendix D. The number of houses to which they were given in each district is shown in Abstract No. 146 on the opposite page; but the figures are somewhat misleading, as in some districts the schedules given to police posts and the like have, contrary to my express instructions, been included in the return.

I found that Deputy Commissioners had not in all cases observed the direction to prepare lists of the houses for which private schedules were sanctioned at an early stage of the operations. This was a mistake; as till the enumerator has an extract from this list for his block, he cannot tell what houses he is to exclude from his enumeration. The schedules were distributed two or three days before the Census night, the headings having been first filled up in the district office. On the morning after the Census the enumerator collected all the private schedules for his block, carefully examining and correcting and completing the entries in each. He then stitched them into the end of his book, and the supervisor, after testing and signing them, added the number of new pages on the outside label. It was originally proposed that the householder should enter in his private schedule all persons residing in his compound. But at a late stage of the proceedings I succeeded in procuring sanction to the rule that he should enter only the residents of his bungalow, the native servants being enumerated in their outhouses by the ordinary enumerator of the block. This was, I am convinced, the right rule to work upon. It would be hopeless to expect an officer of a Queen's regiment or a European tradesman to enter correctly the particulars regarding his native servants. But of course the distinction should in future be drawn in the first instructions issued to enumerators. Several Deputy Commissioners bear testimony to the good effect which the rule had upon the correctness with which the schedules were filled up, and point out that the chief difficulty experienced was where officers in camp, for instance, had to fill up the schedules for Natives as well as Europeans. In many cases these had to be returned for correction or completion.

[1] Section 9, General Letter A, and Rule 6, Part I, Instructions to Supervisors, Appendix D.

[2] Section 11, General Letter B, No. 257 of 16th November 1880, Section 11, Part I, Instructions to Supervisors, Appendix D.

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Part III.—Special Measures.

Abstract No. 146, *showing* issue of Private Schedules *for* Districts.

1	2	3	4	5	1	2	3	4	5
Serial No.	Name of District.	No. of Houses to which private schedules were given.			Serial No.	Name of District.	No. of Houses to which private schedules were given.		
		English.	Persian.	Total.			English.	Persian.	Total.
1	Dehli	59	1	60	19	Rawalpindi	291	106	397
2	Gurgaon	14	13	27	20	Jahlam	40	9	49
3	Karnal	15	7	22	21	Gujrat	6	142	148
					22	Shahpur	7	2	9
4	Hissar	14	48	62					
5	Rohtak	10	30	40					
6	Sirsa	10	5	15	23	Multan	112	21	133
					24	Jhang	7	12	19
7	Ambala	196	...	196	25	Montgomery	7	6	13
8	Ludhianá	17	43	60	26	Muzaffargarh	7	19	26
9	Simla	206	...	206					
					27	Derah Ismail Khan	33	1	34
10	Jalandhar	60	12	72	28	Derah GhaziKhan	37	138	175
11	Hushyarpur	9	27	36	29	Bannu	29	33	62
12	Kangra	70	30	100					
					30	Peshawar	155	4	159
13	Amritsar	64	224	288	31	Hazara	46	85	131
14	Gurdaspur	71	58	129	32	Kohat	23	44	67
15	Sialkot	71	31	102					
						Total	1,915	1,263	3,178
16	Lahore	122	70	192					
17	Gujranwala	16	2	18					
18	Firozpur	91	40	131					

873. The private schedules issued to natives appear to have been as a rule correctly filled up. And it is curious how little desire seems to have been felt to claim the privilege of self-enumeration. In the whole of Dehli only one private schedule was asked for or granted to a native. In Lúdhiánah private schedules were offered to several natives and refused by them, as they preferred the enumeration to be made in the ordinary way; and in Gurdáspur these schedules were almost wholly confined to the district head-quarters, "where the native officials encouraged applicants." On the frontier, where the *pardah* feeling is exceptionally strong, one would have thought that the privilege would have been eagerly sought after. But in Hazára the Deputy Commissioner notes that the schedules arrived somewhat late, and meanwhile many of the houses had been censused in the ordinary way by enumerators without any sort of difficulty or objection. Mr. Dames of Derah Ismáíl Khán writes :—

"Private schedules were issued to European residents and to officers in camp. But few were issued to Natives as they expressed no desire for them. Most of the leading families reside in Derah Ismáíl Khán, and as they are well represented on the committee they preferred to have their Census taken through it, which was done without interfering with their privacy. One was issued to the household of the Nawáb of Tánk and one to Nawáb Alla-Dád Khán, Extra Assistant Commissioner at Bhakkar."

and Mr. Lawrence of Pesháwar says :—

"It was determined not to issue private schedules to any one in the district. We had taken measures to appoint men of family to act as enumerators for villages in which the Arbábs and Kháns reside, and the enumeration took place without any obstruction. In this district it would have been very hard to draw the line, and much discontent and ill-feeling would have resulted from using private schedules."

It almost seems that there would be no difficulty in refusing private schedules altogether to natives if thought advisable. At the same time the Deputy Commissioner of Síálkot is of opinion that their employment "lightened the work;" and Mr. Hawkins of Amritsar writes "the grant of private schedules saved "the establishment, and the entries made were as a rule correct. They might have been more largely "used with advantage."

But the English schedules were as a rule very badly filled up, especially by military and non-official Europeans. The Pesháwar authorities report most unfavourably of the inaccuracies which characterised these returns. Colonel Jenkins of Kángra writes :—

"The same cannot be said regarding the English schedules, the majority of which were wrongly filled in. At least this was the case regarding the schedules distributed in the Dharmsála Civil Lines and Cantonments, which were collected by Mr. Tucker. "It is perhaps to be regretted that those distributed to the planters scattered over the valley were not subjected to some sort of "scrutiny. It would have been easy for them to have been sent to the Sadr, and no great delay would have been occasioned thereby. Indeed though the greater part of the arrangements for the Census have been so elaborate and excellent, there appears to be considerable room for improvement with respect to Europeans.

"It must be remarked that the distribution of private schedules in no way lightened the amount of labour, but that the contrary was the case."

and Major Nisbet of Simla says :—

"In taking the Census of a considerable town like Simla, and that of the non-official European and Eurasian population of the hill cantonments of Kasauli, Dagshai, and Sabáthu, the number of private schedules issued no doubt appears very small. It will be remembered, however, that a Census of the town of Simla was taken so recently as 20th September 1878; and profiting by the experience then gained, I felt it was useless to trust many Europeans and Eurasians to furnish correct Census returns, if private schedules were made over to them to fill up. In the case therefore of a majority of families, the Census returns were filled up and prepared for them on the night of the Census by English-speaking enumerators specially engaged for the purpose."

It is, however, absolutely necessary that private schedules should be used for a large portion of the European population, as native enumerators would be treated by them with neglect, if with nothing worse. I shall presently make some suggestions on this subject.

874. Officers in camp.—Officials who were in camp on the night of the Census enumerated the whole of their camps in private schedules obtained from the nearest Deputy Commissioner. Their camps were held for Census purposes to be included in the block in which their head-quarters lay; and where two officers had a joint camp, the head-quarters of the senior officer were taken for this purpose.

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Part III.—Special Measures.

875. European Census in general.—I believe it is the experience of all Indian Censuses that the Census of the European community is much worse done than that of any other portion of the population, and certainly the Panjáb Census of 1881 was no exception to the rule. The mere enumeration was no doubt accurate enough. But it appears hopeless to expect the average European house-holder to read the instructions for filling up the schedules; and the proportion of Native to European population in any given block is so overwhelming that the enumerators and supervisors are necessarily chosen for their knowledge of the natives.

I examined every entry in every European schedule during the course of tabulation. I found few instances of silly or wilfully incorrect entries. But I found—

(1) that the sect, and the fact of being a European British subject were very generally omitted:
(2) that the nationality was often not rightly entered:
(3) that the name of some obscure English village was often given in the birthplace column:
(4) that the occupation was very generally insufficiently described.

The schedules returned by women were generally far more accurately filled up than those prepared by men. The ages of women were of course stated incorrectly. One lady returned her age as "over 30." I have no reason whatever to doubt the correctness of the entry; but it was not so precise as I could have wished.

I do not think it would be impossible to have, as suggested by Colonel Jenkins (section 873), all the European schedules examined immediately after the Census by a properly qualified European officer, and returned for correction where necessary. The largest number of Christians other than Native Christians in any single district is 4,018, and the greater part of these are soldiers, whose schedules can be run through very rapidly.

6		7	
Religion	Sect	Nationality	European British Subject.

I would also slightly modify the schedule for Europeans and Eurasians, substituting for columns 6 and 7 the columns shown in margin. As the Government of India schedule stood, the nationality and the fact of being a Eurasian or a European British Subject were to be entered in the birthplace column, which not unnaturally confused such people as read the foot-notes directing the entry.

876. Railway Census.—All Railway premises were from the first excluded from the block-system. It would I think have been better to have treated and entered the Railway premises in each Tahsíl as a separate circle, for the enumeration of which the Railway authorities were responsible. The Railway Census was divided into two distinct portions (1) that of the resident Railway population; (2) that of passengers by rail on the night of the Census. The rules on the subject will be found in Appendix D.

The Railway premises, for the Census of all persons residing within which the Railway authorities were responsible, included the roadway, all stations and signallers' and gatesmen's huts, all large quarters occupied solely by Railway employés such as exist at Lahore and Philaur, and all colonies which owe their existence entirely to the railway, like those at Adamwahán and Sher Sháh Bandar. In some cases it was found more convenient to enumerate isolated signallers and gatemen with the village near which their huts were situated; but in many parts, and especially where the Multán line passes through the *bár*, the distances were too great for this to be possible, and these men and their families were enumerated by an official who trollied down the line on the day of the Census; and they were assumed to be present at night.

The scheme followed was identical with that in general use, save that no house-lists were prepared. They were omitted under orders. But I think it was a mistake, as the Railway quarters at Lahore for instance include a very considerable number of houses; and in such places at any rate they should have been drawn up. Each station was treated as a block for the enumeration of which the station-master was made responsible, and at small stations a house-list was undoubtedly unnecessary. At large stations the premises were divided off into blocks, the responsibility for which was distributed among the various Departments. The schedules of each block, when completely tested after the Census, were sent to the Deputy Commissioner of the district in which the block lay.

The passenger Census was conducted on this wise. Between 9° P.M. on the 17th and 6° A.M. on the 18th, each traveller by rail was enumerated at the station at which he alighted. On the morning of the 18th all passengers found in the train, and who did not appear to have been already enumerated, were counted at the first considerable station at which the train stopped at or about 6° A.M. Private schedules were used for first and second class passengers, and for Europeans travelling third class. The few passengers who alighted at wayside stations were easily dealt with by the local staff. Where trains arrived at a terminus during the night, the clerks employed at the terminus during the day were kept on the spot to take the Census. Where the six o'clock Census had to be taken at a station which was not a terminus, sufficient clerks were sent there for the purpose. I saw the Census of the 11° mail taken at Lahore. Pairs of clerks, accompanied by a porter bearing a lamp, were standing on the platform as the train drew up. Each pair took a third class compartment, one taking one side and one the other, and beginning at opposite ends. As soon as they had finished their carriage its occupants were allowed to get out; and the clerks moved down the train to the nearest carriage which had no enumerators at it. The door of each carriage was kept locked till its enumerators were ready to begin work. In this way the Census of the passengers was taken quickly and well. The passenger schedules were kept entirely separate from those of the resident railway population, and were treated separately at tabulation.

877. The instructions for the Railway Census were not issued by the Government of India till the end of December 1880, which left us very scant time to make the necessary arrangements; and I did not hear of the completion of the preliminary record for the whole Panjáb railway population till the day of the Census itself. The arrangements on the Scinde, Panjáb and Dehli and Indus Valley Railways were admirable, and the Census of the small part of the Rájpútána State Railway lying within Panjáb limits was

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Part I.—Abstracting the Figures.

officers were located at the head-quarters of divisions except those of Pesháwar, Deraját, and Hissar, which were stationed respectively at Ráwalpindi, Multán, and Dehli so as to be easily accessible by rail[1].

893. I am inclined to think that, on the whole, the plan adopted was the best. The single central office is, I am convinced, quite out of the question for the Panjáb. The only staff that could hope to deal successfully with the infinite variety of local terms is a local staff such as we employed. Take a single instance—that of occupation. *Faqír* means a tenant in Pesháwar, a beggar elsewhere. *Daftari* means an office menial save in Pesháwar, where it is a landed proprietor. *Páli* and *Banjárá* mean respectively a herdsman and a cattle merchant in the east, an oil-presser and a pedlar in the west, and so on *ad infinitum ;* while the local terms which are absolutely unintelligible to any outsider are simply innumerable. So, too, we found local knowledge invaluable in the grouping of blocks and hamlets under the proper villages and towns, and of villages in their proper places ; though if the tahsíl abstracts had been properly prepared this should have presented no difficulty. But the necessity for local knowledge is felt at every stage of this portion of the work.

The choice then lies between district and divisional tabulation. I do not attach much weight to the objection that uniformity would be impossible if each district tabulated its own figures. So long as the operations are confined to *tabulating*, not *classifying*, the various entries in the schedules, the process is purely mechanical, and uniformity is only convenient because it is well that the easiest and best method should be followed everywhere. A scheme would be drawn out beforehand ; and a good Tahsíldár would do the work more quickly, better, with a smaller expenditure of power, and with infinitely less hindrance to current work than would be possible were it done in a divisional office. But there are bad Tahsíldárs, and those not a few. It would be difficult for the district staff, already burdened with arrears of work accumulated during Census operations, to exercise the continual supervision which alone would be effective ; it would be impossible to put a superior officer on special duty for each district ; and though in a large portion of the Province the work would be admirably done, yet bad work would be an absolute certainty from some tahsíls.

The two dangers are delay and fudging. The first should be easily avoidable. We took three and a half months abstracting and tabulating with one-eighth of the Provincial revenue establishment ; and if we had tabulated for tahsíls only instead of for districts and divisions also, we should have saved three weeks out of that. The work could be done in the tahsíls in a month or six weeks with the greatest ease. But the danger of fudging would be infinitely greater. As I have noticed in section 891, the strain of Census operations once relaxed, there is a tendency to scamp further work done in the districts ; while in the divisional offices the staff take a fresh departure. It is true that the inconvenience and discomfort entailed upon the patwáris by living away from their district for three months is so great that there is a strong temptation to make the figures agree by any means, and so have done with them. But native clerks will fudge under all circumstances if they can ; and everything depends upon the supervision. Now it is possible to select ten first-class supervisors for the divisional offices ; but it is not possible to procure thirty-two such men for the districts. Thus, on the ground that there are some Tahsíldárs under whom the work would certainly be fudged, and on that ground alone, I am inclined to think that the divisional offices offer the best practicable solution of the problem. The consideration that weighs most strongly with me on the other side of the question is, that if the tabulation were done in the districts, errors detected in the tables would be sent to the districts to be put right ; and the small compiling staff would not be, as it now was, overburdened with the labour of correction. But I think that the burden might in future be largely reduced by measures to be described later on, and in any case the difficulty could be met by increasing the compiling establishment.

894. Preliminary examination of schedules.—When the schedules reached the divisional offices, they were examined in order to make sure—

(1) that we had a book of schedules for every block in the Province ;
(2) that the series of pages in each book was complete, and that their number was not less than the number of pages entered on the outside label ;
(3) that the series of house-numbers was complete, and that each house had an entry of some sort against it, even if only to state that it was empty ;
(4) that the private schedule for every house which was noted as being allowed such a schedule was stitched into the book.

Where the schedules were found defective in any of these respects the Tahsíldár was referred to for an explanation or requested to send the missing papers. This examination assured us that we had a Census entry for every house entered in any house-list throughout the Province. Having this assurance, we assumed that those entries were correct save in the very few cases noted in section 899. Some of the offices began making a more minute examination ; for instance, to ascertain whether all the families and heads of families shown in the house-list were entered in the schedules. But I put a stop to this. The house-list was prepared some months before the final Census ; it was quite possible that the apparent omission, where it occurred, was correct ; and if we had examined every entry to discover whether it afforded room for suspicion of inaccuracy, our labour would have been endless while the errors corrected by it would have been inappreciable in number and importance. I was astonished to find in how extraordinarily few instances did this preliminary examination afford any grounds for inquiry, and how hardly ever did that inquiry disclose any real omission. In tahsíl after tahsíl not a single apparent flaw could be detected ; and the schedule record of houses, at any rate, was practically perfect.

895. The process of abstracting.—The abstracting was done by abstractors who took out the figures

[1] Probably by next Census the facilities for communication and even the district and divisional boundaries will have altered considerably. But I am not at all sure that it would not be better to tabulate the figures for the Bannu district at Pesháwar and not at Multan not only because the distance is smaller, but also because it is a Pathán and Pashto-speaking district.

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Part I.—Abstracting the Figures.

from the schedules and entered them in registers devised for the purpose. These registers were eight in number, and showed the following details in addition to the sex, which was distinguished in each :—

1. Age. Civil Condition. Religion.
2. Musálman Sects.
3. Birth-place. Language.
4. Caste. Certain Religions.
5. Occupation, under or over 15 years of age.
6. Education (including uneducated). Age. Religion.
7. Infirmities (four kinds). Age. Religion.
8. Sects and Races of Christians.

The original plan of working was this. Four men took a book of schedules: blank registers were distributed among them: one man read out the entries opposite each name, and as each entry was read the clerk in charge of the register in which that entry had to be abstracted made an upright stroke in the appropriate column. Each fifth stroke was drawn diagonally across the preceding four, so that they were grouped into fives and could easily be counted.

I do not propose to enter into minute details of the method of working. But there are one or two matters of principle which it will be necessary to discuss here. Some of the registers were very complicated; for instance, the one which showed civil condition and age had the age-periods printed down the side and the form ruled off into spaces for the various religions, each of which was again divided into three spaces, *single*, *married*, *widowed*, under each of which were two columns for *male* and *female*. Thus the clerk, on hearing the entry "married, male, 9 years, Hindu," had first to find the horizontal line on which entries for the age-period "5 to 9" were to be made, then to follow that line till he came to the space for Hindus, under that to select the column for married, and in that the sub-column for males. There he made a tick. In filling up registers of this complicated nature it was found indispensable that the entries should be read out; as if a single clerk attempted to abstract straight from the book, he lost his place in the register every time he looked at the book, and delay and error resulted. But in simple registers such as that for Musalmán sects, where the clerk had only to distinguish sex and five sects, it was far quicker and easier for him to abstract straight from the book. Now in the Hissár division, out of a total Musalmán population of 286,316, no less than 285,668 are Sunnis: that is to say that, probably in 99 out of every 100 blocks, every single soul was a Sunni. But a block containing 499 Sunnis and 1 Shíah could be abstracted in two ways; the clerk could either make a tick for each entry, that is to say 499 ticks in the Sunni and 1 in the Shíah column; or, he could run his eye down the column for sect, make a tick for the one Shíah, and find the number of Sunnis by subtracting him from the 500 Musalmáns which he would find entered in the religion register. Exactly the same two methods were possible with birth-place, language, education, and infirmities, in all of which the enormous majority of entries in each block are one and the same, and the exceptions can easily be picked out.

Mr. Elliott permitted either of these two plans to be followed, and, availing myself of this license, I largely followed the latter. I have since had reason to believe that Mr. Plowden objected to its adoption on the ground that it afforded no complete check. This is true. If there were two Shíahs in the block and one of them escaped the eye of the clerk, the result would not show that there was an error; whereas if he made a tick for each entry, it is less probable that he would make a tick under Sunni for the second Shíah; and unless he did this the total would be wrong and the figures would be re-examined. For similar reasons Mr. Plowden insisted upon our abstracting figures both for the educated and for the uneducated separately for each age-period used in the age-register; not because the details of age were needed in combination with education, for they were not even tabulated; or because the details of uneducated were of any use; but because, by abstracting the details of age twice over for the whole population independently, we obtained a check upon the abstracting, as the two results had to agree or the work was done over again.

896. The system of check.—And this leads to the whole question of check on the abstracting. The checking staff was, of course, kept quite separate from the abstracting staff, the work done one day by the latter being handed over next morning to the former to be checked. Here the totals and details of the different registers were compared with one another and with the figures of the village abstracts (which last were never allowed to pass into the hands of the abstracting department at all), and if any differences were detected the figures which were doubtful were re-abstracted; while even if there were no discrepancies, a certain proportion of the figures were tested by being got out again by the checkers, the proportion varying with the degree of accuracy with which the men whose work was under examination were found to abstract. Now my first rule was that under no circumstances whatever was any preliminary or intermediate check to be allowed; that the registers were to be totalled in the checking, and not in the abstracting department; and that till they passed into the hands of the checker, no examination of any sort was to be made. This practically amounted to saying to four clerks who sat down together to abstract the figures of a single book: "You must get out your figures independently of each other, and must not examine "them to see if they agree with one another. I will do that for you; and if they do not agree you will "have to do them again, and will be punished besides." The plan was admirable in theory; but I ought to have known that, with natives at any rate, it could not be worked in practice. No supervision could prevent the men comparing their figures as they went along, and I was compelled to recognise and regulate the practice. If the men found, at the end of abstracting a block, that one register gave 253 and 219 and the other 252 and 220 for Hindus and Musalmáns respectively, the discrepancy could be removed either by re-counting the Musalmáns, or by assuming that one register was right and altering the other to agree with it. The former method gave the correct results; the latter was fudging. What I had to aim at was to make first the honest correction so easy, and secondly detection of the dishonest correction so probable, that it was not worth while to fudge.

Part I.—Abstracting the Figures.

The first end I attempted to attain by making them compare the figures for each opening of the book—say 25 to 50 souls—as soon as it was abstracted. If any detail did not agree, it was almost as easy to run the eye up the column and get the right figures for the detail in question, as to fudge the corrections so as not in the act of doing so to produce a second discrepancy. For the detection of fudging (or of honest error) there were three means. First the comparison of the figures of the registers one with another. This only availed when the fudging was clumsily done, and was really nothing more than a very necessary check upon and repetition of the comparison already made by the abstractors. Secondly the comparison of the abstracted figures with the figures of the village abstracts. This was certainly a check in that it often led to re-examination of the figures; but it extended to only a few totals and really gave us much unnecessary labour, for in the enormous majority of the cases of difference, the village abstracts were wrong and the register right. Thirdly the testing the work by taking a certain number of items from each register and getting out the figures for them from the schedules independently. This was the real safeguard; and upon the efficiency with which this test was applied and the extent to which it was carried depended the measure of confidence which could be placed in the accuracy of the results.

But it is evident that the allowing the abstractors to compare the figures as they went along very greatly reduced the value of any inter-check between the registers, for they were practically no longer independent. If the clerk who left out a Shíah in section 895 found his Musalmáns short by one, he might look at the book again and detect his omitted Shíah, or he might save himself the trouble by adding on a Sunni; but in no case would the register show that there had been a discrepancy. Thus it was almost as safe to allow him to pick out his Shíah at once, and take the rest as Sunnis without counting them. So, too, it made the combination of age details with education of very small value as a check; for though the existence of the two registers showing age would perhaps disclose an error if it had occurred, yet it disclosed it to the abstractor, and the only way of finding out whether he had corrected it or fudged it was to re-abstract the figures. And it made the retention of the working party of four independent abstractors unnecessary, if any speedier method could be devised.

897. Proposed system of abstracting.—We tried many ways of combining our clerks and registers, for I allowed considerable license in this matter within the limits which I considered necessary to ensure accuracy, and which I insisted upon being strictly observed; as for instance, when I directed that the age registers were in all cases to be filled up from dictation and not by a single clerk. I found that men got accustomed some to one method and some to another, each preferring his own; and above all that certain men worked well together in parties of two, three, or four as the case might be, and gave the largest and best outturn when working in their own fashion. A slow or stupid man was a terrible drag upon sharp companions, who could really turn out more work without him. But on the whole the following is the plan which I believe to be the best, and which I would recommend for future adoption. Two clerks take two books of schedules; one reads out the entries while the other fills up the register for age, religion, and civil condition. Then each takes a book and fills up the entries for education, infirmities, sect, language, and birthplace, only counting those who are educated or infirm, or who belong to a strange sect, speak a strange language, or were born out of the district. In the same way each man fills up the registers for caste and occupation. It is true that in these two last registers *every* caste and occupation has to be entered. But they are nevertheless better filled up by one man than from dictation, as the caste and occupation is very generally constant for each family, and often for a house or even a group of houses. The clerk notes how far down the page the entry is the same, counts up the items he wants within those limits, and enters them at once. The education register we had to prepare from dictation because we were compelled to take out age statistics in it. But this I think was unnecessary.

The system I have sketched above lessens the value of the check afforded by comparison of registers one with another, because it does away with the working party of four independent abstractors. But as I have already pointed out, the value of that check is almost destroyed by intermediate comparison, which as I have explained above it is quite impossible to prevent. I am convinced that the only really effective check is the examination of a percentage of all work done by doing it over again; and during the earlier stages of the work, before the men become expert, the percentage so examined should be large. After a time it is marvellous how accurate the work becomes, and how rarely an error occurs. There is, of course, one way in which absolute accuracy can be secured. That is, by having each register prepared perfectly independently and by a separate staff of abstractors for all the blocks, and not allowing the staff who are abstracting one set of figures to have any knowledge of the results obtained by those who have abstracted another set. The registers would then be compared; and if two sets of figures did not agree they would again be abstracted by two new sets of men who had no knowledge of the figures obtained in the first abstraction. There are many practical objections to this system. There would, I think, be danger of confusion, delay, loss of papers, and omission of portions of the process. And I doubt whether, with native agency, it would be possible to prevent the results being communicated for the purpose of fudging. But an absolute fatal objection, to my mind, is the enormous expense attending it. I understand that it has been followed at Madras, and has been so costly that the Imperial Government have declined to pay for it.

898. Cost versus accuracy.—And here I would fain deliver my soul as to the ethics of fudging. The practical question that has to be considered in all operations of this sort is, what is the attainment of any particular degree of accuracy worth; given that a certain amount of extra labour and extra check will secure a certain enhanced measure of accuracy, is the increase in accuracy worth the cost of attaining it. And I think one very obvious limit is afforded by the consideration, that under no circumstances can it be worth while to tabulate material with a degree of accuracy greater than that which the material itself may reasonably be supposed to possess in the first instance. I have heard that in a certain Census the whole staff was engaged for six weeks in examining and removing a discrepancy of *four* in the total population of a Province. There were special circumstances which justified the delay in this case; but under ordinary circumstances I should look upon it as waste of time, and should consider conscientious fudging

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Part II.—Tabulation of the Figures.

but at the expense of accuracy, and his tables were sent in in a state of disgraceful incompleteness. All errors in tabulating and totalling were detected and removed in the central compilation office. But cases in which separate occupations or castes had been wrongly included under one heading unfortunately could not be detected till the results in their final form came to be discussed; and it was then as a rule too late to alter them (see further, section 918), though we did in the worst instances go back to the original schedules, at the cost of great time and labour.

908. But it must be remembered that this Census was, as far as the Panjáb is concerned, practically a first experiment. Nothing on the same scale or of the same degree of minute and accurate classification had been attempted before, and it was of the necessity of the case that errors should occur and mistakes be made. Those errors and mistakes I have endeavoured to set forth faithfully, not only in order to show how far the figures really are what they appear to be, but also in order that at a future Census their repetition may be guarded against. It was proposed by the Government of India that a list of castes and occupations should be prepared before the Census for the use of the enumerating staff. Anything more fatal to accuracy and fulness of detail I can hardly conceive. It would have been wholly impossible to prepare even approximately complete lists, or to use them if prepared, for they would have been of perfectly impracticable bulk. Moreover, to put a list of the sort into the hands of the enumerators would, where it did not simply bewilder them, cause them to enter, not what the people told them, but what a very cursory examination of the list might suggest as being somewhere about the truth. So, too, a native who, if asked what his clan was, would give a perfectly correct answer, would, if asked to which of a certain number of specified clans he belonged, almost certainly name one of them, though he might really belong to none. And it would be fatal to tell the enumerator to record clans in some instances and not in others. The only safe method is that adopted at the present Census, *viz.*, to *record* everything, leaving the process of selection and rejection for the after-stages of the work where the agency is of superior quality, supervision is closer, and uniformity can be secured. Moreover, while fairly complete classified lists would have been of immense value in the compiling offices, it would have been, as the Panjáb Government pointed out, quite beyond the bounds of possibility to prepare such lists; for not even the merest rudiments of the necessary material were in existence, while the Census itself afforded the only possible means of obtaining it. The result has shown how correct this opinion was; far more so probably than any of us conceived at the time.

At a future Census the case will be very different. The lists of castes and occupations included under each heading of the tables, to which I have already referred as having been prepared in the divisional offices, will, taken together with the table-headings themselves and the MS. tables of tribal sub-divisions, form a very fairly complete detail of the entries in the schedules: and this material renders possible the preparation of classified lists for use in future compilation. The Panjáb Government has approved of my proposal to devote my leisure for the next few years to the preparation of such lists; and I hope that I shall be able to complete them, and that they will be found useful. But I am convinced that the only stage in which they can legitimately be used is in the process of tabulation and compilation, and that any attempt to use them for guidance in enumeration will be fatal to accuracy. We cannot record facts at enumeration in too minute detail; we can classify and compress afterwards. To attempt to classify in the first instance through a widely-spread agency of inferior average qualifications and subject to no central supervision might perhaps tend to uniformity, but it would be a uniformity of error.

909. Units of tabulation—Towns and Villages.—Perhaps no single point throughout the Census has given me so much trouble in proportion to its importance, as the distinction between urban and rural population. There were two questions to be decided—

(1) What places should be classed as towns:

(2) What limits should be taken for the urban area in each place.

The correspondence and instructions issued on the subject will be found in Appendix D. The conclusions finally arrived at were—

(1) that all municipalities, civil lines, and cantonments were to be treated as towns, whatever their population:

(2) that all places including a population of as many as 5,000 souls dwelling together within a practically conterminous collection of houses and suburbs should be classed as towns:

(3) that places not falling within either of the above classes, but the population of which was urban rather than rural, not essentially consisting merely of a village community with its attendant menials and traders, but also including a considerable extraneous society drawn to the spot by the presence of a shrine of peculiar sanctity, by the existence of a centre of local commerce, or the like, should be reported for orders.

A good many cases were reported under clause (3) above, in pursuance both of my General Letter A and of my letter of 1881. But in all such cases the Panjáb Government declined to depart from the numerical standard of 5,000 souls. I myself should have been inclined to consider the circumstances in some of these cases sufficiently special to justify an exception, if any exception was to be made at all. But I would, on the whole, make no exception. It may be said generally that, for reasons explained in detail in sections 62, 63, it is often difficult to decide in respect of any given place whether its population is rather urban or rural in its composition; and an arbitrary numerical standard seems to be both necessary and unobjectionable.

With respect to the limits of urban areas, the cantonment boundaries were followed in the case of cantonments; while the Deputy Commissioner fixed limits for civil lines where not already included within the boundaries of a town. In the case of municipalities the municipal boundaries were followed in all cases except that of Lahore, where those boundaries included several villages purely rural in their nature and lying at a very considerable distance from the town, and which were excluded from the urban population. A detail of them will be found in the Appendix to Table XIX. There is no doubt that Lahore is not the only case in which the municipal limits include small villages which would, *per se*, be better included

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Part II.—Tabulation of the Figures.

in the rural population. But it is so important to know exactly what has and what has not been included within each town, that it is I think right, save under very special circumstances, to follow the only boundaries available which are capable of exact ascertainment. The point was specially referred to the Deputy Commissioners for opinion with reference to each town in the Province; and the only case in which exception was thought advisable was Lahore. As already pointed out in section 845 the details of suburbs were, at all events as a rule, distinguished in our tables.

920. **Units of tabulation—village boundaries.**—In the case of all places not being cantonments, civil lines, or municipalities, the boundary of the village area was taken as the limit of the unit of tabulation, whether classed as town or village. There is no doubt that, as explained in sections 65, 65, these limits are in some degree and in some parts of the Province misleading as regards the size of the village. But no other limits would have been practically possible. In the first place it was absolutely essential that the tables should be based upon the *mauzahwár* lists of village areas which are kept at the revenue record-rooms; and this, not only to ensure that no portion of the Province was omitted, but also to prevent the recurrence of an error such as occurred in the Census of 1868, when a wholly imaginary town of the name of Májra, with a supposed population of 7,908, was created in the Rohtak district by adding together the small populations of the numerous outlying hamlets which are commonly known to the people as *májras* in that part of the Panjáb. In the second place the villages of the Province present every gradation of compactness or the reverse, from the single homestead of the Dehli territory, with its outlying temple, monastery, perhaps some gypsies' or cattle-traders' huts, and occasionally a hamlet or even two containing 10 to 20 houses each, to the *tappa* of Kángra and the *kothi* of Kúlu where each family inhabits a separate hamlet, or the *mauzah* of parts of Multán and the Derájat where the population is scattered all over the area at isolated wells or on lowlying plots of cultivation. Now in the former class the outlying population of say 4 to 80 souls are necessarily included with the parent homestead. In the latter class there is no alternative between either treating the village as a whole, or tabulating separately countless little hamlets, each with a population of from 20 to 100. In all cases it would be wholly impossible to draw any line which could be worked upon in practice. In the hill districts, indeed, there are well-recognised and clearly-defined sub-divisions of the *tappa* and *kothi*, known respectively as *tikas* and *phátis*: and these might perhaps prove more convenient units of tabulation than the larger areas. But Mr. Lyall tells me that in most cases they would be impracticably small. It has been suggested that an arbitrary limit of say 500 souls should be adopted, and that all hamlets reaching this size should be tabulated separately. But, besides the other objections just stated, there is the, as it seems to me, fatal obstacle that it would be necessary to tabulate all hamlets separately before it could be discovered which of them reached the limit in question. At the same time the information if it could be obtained would be most interesting, as it would gauge the progress of the process described in section 170. At present our figures for villages are almost meaningless in some parts of the Province.

The village area being accepted as the unit, it became important to exclude from the list of towns all villages, such as those of Derah Ismáíl Khán mentioned in paragraph 66, in which the population, though exceeding 5,000 in number, was widely scattered over a considerable area. Here again every degree of diversity existed, the principal homestead containing any number from 400 to 4,000 souls, and the remaining population lying at various degrees of proximity to it. The case was one which only local knowledge could deal with; and lists of doubtful cases were sent together with a copy of the correspondence on the subject to district officers, who decided in each case whether the population was or was not sufficiently compact for the place to be treated as a town. Table XIX shows 132 towns of over 5,000 inhabitants: Table XVIII shows 164 towns and villages of that size in British Territory. The difference consists of 32 villages excluded from the list of towns on the grounds just stated, in the following districts:—Jálandhar 1, Kángra 13, Derah Ismáíl 6, Derah Gházi 4, Bannu 6, Hazára 2.

921. **Towns and villages—precautions adopted and suggested.**—Such were the principles on which the classification was made. But they were very difficult to apply in practice. As already noticed in section 845, my instructions regarding urban areas had not been understood, and the schedules of suburbs which lay within municipal limits had nothing to show that they properly belonged to any town. The first thing therefore was to prepare, with the help of the tahsíl abstracts and the local knowledge of the staff, lists of all places apparently falling within the definition of town, showing all the component parts of each and their populations. These were sent to district officers with a copy of the correspondence on the subject, and they were asked to report—

(*a*) whether any places had been included which should not be classed as towns:

(*b*) whether any places had been omitted which should be classed as towns:

(*c*) whether all suburbs or outlying homesteads situated within municipal limits or the boundaries of cantonments or civil lines had been included:

(*d*) whether any of the subsidiary parts shown lay outside those limits or boundaries:

(*e*) whether any of the subsidiary parts lying within municipal limits lay so far from the town that it would be well to exclude them on special grounds.

If the answers to these inquiries had been absolutely accurate, my table of towns would have been perfect. But I had occasion later on, when the final figures were past alteration, to circulate a rough copy of the Appendix to Table XIX in print in order to have the spelling of names of suburbs corrected; and it then became apparent that in two or three cases, which an examination of column 4 of the Appendix will disclose, the answers had been inaccurate, and that I had consequently included certain small suburbs which lay beyond and excluded others which lay within municipal limits. And I cannot feel sure that we may not in other cases also have omitted suburbs which ought to have been included; for in the index-register and tahsíl abstracts, which were all we had to guide us, suburbs were sometimes shown as independent villages; and the fact that only in two or three cases did Deputy Commissioners report that we had omitted suburbs proves either that our local knowledge was singularly sufficient, or that enough care was not taken in examining our lists. In any case, however, the errors thus caused will have been comparatively insignificant in amount; for no suburb of any practical importance can have escaped inclusion with the town.

Part II.—Tabulation of the Figures.

At a future Census the plan of operations I would recommend would be something of this sort. The urban limits would be governed in all cases by the considerations just discussed. They would then be split up into sections and blocks in the manner described in section 846. The Deputy Commissioner and one of his Assistants would naturally visit every town in the district during Census operations, and would then examine the manner in which this work had been done on the spot. If he found reason to think that municipal limits should be departed from, he would report his proposals; but I think very strong grounds should be shown before this was allowed. When it was finally settled what was to be included in each town, special labels of a different colour from those used for rural books of schedules would be pasted on the urban books, in which, instead of the name of the village, the names of the (1) town (2) suburb or other sub-division should be shown separately. And it would help the compiling offices if the bindings of urban books were also covered with cloth of a different colour from that used for the rural blocks. The index-register of towns would be kept separate from that of villages; and in it care would be taken to show the name of the town in column 3 and that of the suburb in column 6 only; while in the tahsíl abstract for towns, which should be separate from that of villages, there would be two columns, one for the name of the town and another for that of subsidiary parts. These arrangements, however, would apply in general to municipalities, civil lines, and cantonments only. Village areas including a population of 5,000 or more cannot in all cases be picked out till at any rate the preliminary abstracts have been prepared and the population thus ascertained; they seldom possess any suburbs worth separating; the village boundary is followed as a matter of course; and all that is necessary is to exclude cases in which the population is scattered.

912. Floating population.—There was a certain floating population to be dealt with, generally but not always inconsiderable in amount, consisting of passengers by rail and boat, and of large gangs of workmen collected upon canals, railroads under construction, and the like. These were not included in the population of any village or town, but were entered in the tables at the end of the village population of the tahsíl in which they were enumerated. The resident Railway population was of course treated exactly like other residents.

913. European schedules.—The mass of the work being done in the vernacular, I was in considerable doubt as to how to deal with schedules filled up in English. I had at first thought of collecting them all at Lahore and abstracting and tabulating them by a separate English staff. But the divisional officers objected that this would involve two difficulties. In the first place we did not know where all the English schedules were, a considerable number being scattered about through the books of the Province; they would have been constantly turning up in small numbers which would have been very difficult to deal with, while there would have been some danger of loss. And it would have been very difficult to arrange for the incorporation in the figures of the village tables prepared by the Divisional offices, of the details of the English schedules tabulated by a central staff. In almost all the Divisional offices there were one or two men who knew English; and where this was not the case I sanctioned the entertainment of a couple of clerks for the purpose. These men made translations of the English schedules into Hindústáni, and read them out to the abstractors. On my visits of inspection I paid special attention to the subject, and especially to the translation of names of occupations, of which I caused lists to be shown me together with the translations used. But the result was not satisfactory. The age and sex were of course tabulated with ease; but there were difficulties in respect of religion, sect, language, nationality, and birthplace; while the tabulation of occupations was exceedingly unsatisfactory. At a future Census I should prefer to keep the population enumerated in English wholly separate. The great mass of it is European; and so long as the numbers of each sex were communicated to the Divisional offices to be incorporated in the village tables, the exclusion of the other details from those tables would not impair the utility of the figures for general district purposes; while supplementary tables could be supplied to the district office where large European populations existed. I would then have all the English schedules for each district collected together by the Deputy Commissioner in the first instance, each bearing plainly the number of its circle and block and the name of the village or town to which it belonged, and send them all to the central office to be tabulated by English clerks under the direction of the Deputy Superintendent. The figures would be added to those furnished by the Divisional office for each district; while only the number of each sex in each village or town would be communicated to the district offices for incorporation by separate entry in their tables.

914. Tabulation of tribal sub-divisions.—I have already explained (section 888) that we did not re-abstract the figures for tribal and caste sub-divisions, but accepted the figures of the third part of the village abstracts prepared by the enumerators. Nor did we tabulate them for any smaller unit than the tahsíl. The number of entries was so enormous that getting out even this detail involved very great labour. The number of sub-divisions shown in the tables for Jats and Rájpúts in British Territory are given in the margin. No other caste approached these numbers. The method adopted was as follows. A clerk took a bundle of village abstracts, and a number of blank slips having the name of the caste at the top and two columns to the left hand for the names of the two sub-divisions. Taking up Bráhmans for instance, he wrote all sub-divisions beginning with A on one set of slips, those beginning with B on the next, and so on, heading all these slips with the word Bráhman. If in the village abstract he found 5 males entered opposite a certain sub-division,

Number of Tribal Sub-divisions for Jats and Rajputs.

No.	District.	Jats.	Rajputs.	No.	District.	Jats.	Rajputs.
1	Dehli	2,878	722	19	Ráwalpindi	679	1,631
2	Gurgaon	732	433	20	Jahlam	969	813
3	Karnal	1,579	700	21	Gujrat	2,946	506
				22	Shahpur	1,130	2,253
4	Hissar	1,936	847				
5	Rohtak	2,986	590				
6	Sirsa	2,159	798	23	Multan	560	1,437
				24	Jhang	1,813	1,978
7	Ambala	1,863	560	25	Montgomery	1,170	1,403
8	Ludhiána	1,526	513	26	Muzaffargarh	3,734	306
9	Simla	56	186				
10	Jalandhar	398	245	27	D. I. Khan	3,739	149
11	Hushyárpur	337	587	28	D. G. Khan	6,400	329
12	Kángra	274	985	29	Bannu	2,862	486
13	Amritsar	258	225				
14	Gurdáspur	365	417	30	Peshawar	740	300
15	Sialkot	639	381	31	Hazára	109	249
				32	Kohát	380	495
16	Lahore	2,572	1,232				
17	Gujránwála	2,264	621				
18	Firozpur	3,167	1,548		Total	52,380	23,925

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Part I.—Subsidiary Matters.

944. Printing of Forms and Instructions.—The number of forms and instructions printed was as follows:—

Forms printed for the Census of 1881.

		No. of Copies.
1.	Schedules, Persian Character	1,130,200
2.	Ditto Nágri and Gurmukhi do.	198,700
3.	Ditto English do.	20,700
4.	Private Schedules, Urdu	3,500
5.	Ditto English	12,000
6.	Labels for Enumerators' Books	72,000
7.	Boat Tickets	7,000
8.	Railway Check Tickets	27,000
9.	Translation of General Letters (complete)	2,200
10.	Instructions to Supervisors (complete)	12,000
11.	Ditto Enumerators (complete)	72,000
12.	Instructions for Boat Census	2,000
13.	Ditto Military Census	1,000
14.	Sample Village Abstracts	10,500
15.	Blank Ditto	77,500
16.	Registers for Abstracting	900,000
17.	Check Sheets for do.	65,000
18.	Caste and Clan Abstracts and Slips	300,000
19.	Miscellaneous Forms and Slips (say)	10,100
	TOTAL FORMS PRINTED	2,930,000

Of these the only ones of which we had any considerable balance left unused, were No. 2 of which 78,600 were left over, the first indents having been exaggerated, No. 11 of which 5,750 too many were printed, and No. 4, which was hardly used at all. Out of these forms I supplied the Hill States, and also the Railway Census of Sindh and Rájpútána. The Persian and Nágri schedules were printed by steam at Calcutta, the other papers were lithographed at Lahore and Lúdhiána.

Major Grey of Ffrozpur suggests that the headings of the schedules should be lithographed instead of printed, as "few natives can read the printed character." But this would be quite impossible. Nothing but steam printing could have supplied the enormous number of schedules required within a reasonable time. Even were the suggestion that one heading only should be used for each book of schedules adopted (see section 957), the number of headings would still amount to something like 60 000. The heading is soon learnt off by the enumerator, and the printed character is at any rate sufficiently legible by him to save him from mistakes.

The text of the instructions was necessarily lithographed, as printed copies would have been practically useless. The greater portion of this work was done at the Lahore Central Gaol under the superintendence of Dr. Dickson, to whom I express my obligations in another place. But I was frequently compelled to have recourse to local presses, who were under no restrictions as to the number of hands employed or the number of hours during which they could work. Most of the registers for abstracting were printed at these last. The Gurmukhi schedules and instructions were printed for me by the Revd. Mr. Wherry at the American Mission Press, Lúdhián, and to him also I have to tender my most sincere thanks.

The rates charged by the Calcutta press were extraordinarily low; but unfortunately we began work so late in the Panjáb that we were obliged to get all our schedules up from Calcutta by mail train, which enormously increased their cost. The Central Gaol rates were inordinately high; but the money went into the Government pocket, and the only ill effect of the unduly high charges was to make the apparent cost of the Census greater than it should have been.

The paper used throughout for all Census forms was yellow Royal paper from the Bally Paper Mills at Calcutta. It proved exceedingly suitable for the purpose.

The English printing was comparatively small in amount so far as Census operations were concerned. But the printing of the final tables was a very heavy job. It was executed at the Central Gaol Press, and though the delay was excessive and the cost inordinate, the work was fairly well done in the end. The text of the Report was printed at the Government of India Press at Calcutta, and the style of work speaks for itself. I am under the greatest obligations to the Superintendent of Government Printing in India and his staff for the manner in which the task has been executed. The work was difficult, the abstracts required great nicety, the spelling of the vernacular names gave much trouble, my handwriting is singularly illegible, and I am much indebted to the care and skill which have produced so exceedingly satisfactory a result. The printing has taken a very long time; but if I had had to depend upon local presses for the work, the issue of the Report would have been still longer delayed.

945. Distribution of forms and instructions.—The distribution of forms and instructions to the Districts of the Panjáb was undertaken by the Lahore Central Gaol; and I think that this is the best plan that can be suggested. If, however, at a future Census the schedules are printed, as they should be, some months beforehand, they would best be sent direct from Calcutta to the several Districts, the supply needed by each being calculated roughly on the basis of the present expenditure (see section 946), and a surplus being kept at Lahore from which to meet supplementary or emergent indents. We were *most* unfortunate in the matter of distribution. The supply of schedules from Calcutta was very fitful, and not as rapid as I had hoped from the promises made me, and they had consequently to be doled out to Districts as they were received, instead of the whole supply for each being despatched at once. We naturally supplied the most inaccessible Districts first, leaving those on the line of rail to receive their schedules later on. But just as we had sent off the schedules for the Districts on the Panjáb Northern State Railway that line stopped through-booking of goods or parcels, and not only did so, but kept our parcels lying for a fortnight at various stations along the line without giving us notice of the fact. Some of the parcels for the most inaccessible districts miscarried; and the result was that in one or two Districts they were obliged to supplement the supply in a small measure by schedules ruled by hand. So, too, with covering boards for the books, which, as I have already pointed out, should be made in the Districts on a future occasion. Instructions were sent by post, and no difficulty was experienced in regard to them.

946. Consumption of schedules.—The number of schedules supplied to the various Districts is shown below[1]:—

[1] By "schedule" I mean a paper with one top heading on each side of it, forming when doubled across the middle, 4 pages or 2 leaves in the enumerator's book.

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Part I.—Subsidiary Matters.

Consumption of Schedules in each District.

Districts.	No. of Schedules.	Total Population.	No. of Schedules per 1,000 Souls.	Districts.	No. of Schedules.	Total Population.	No. of Schedules per 1,000 Souls.
Dehli	39,500	643,515	61	Ráwalpindi	45,750	820,512	56
Gurgáon	35,000	641,848	55	Jehlam	38,000	589,373	64
Karnál	45,000	622,621	72	Gujrat	42,000	689,115	61
Hissar	32,350	504,183	64	Sháhpur	29,000	421,508	69
Rohtak	37,400	553,609	68	Multán	41,620	551,964	75
Sirsa	17,200	253,275	68	Jhang	21,500	395,296	54
Ambála	83,500	1,067,263	78	Montgomery	21,500	426,529	50
Ludhiána	45,000	618,835	73	Muzaffargarh	20,000	338,605	59
Simla and Simla States	5,500	493,314	112	Derah Ismáíl Khán	24,150	441,649	55
Jalandhar	55,100	789,555	70	Derah Gházi Khan	27,520	363,346	76
Hushyárpur	69,300	901,381	77	Bannu	28,180	332,577	85
Kángra	69,230	730,845	95	Pesháwar	34,200	592,674	58
Amritsar	55,540	893,266	62	Hazára	35,180	407,075	86
Gurdáspur	54,520	823,695	66	Kohát	16,200	181,540	89
Siálkot	56,100	1,012,148	55				
Lahore	47,700	924,106	52				
Gujránwála	43,500	616,892	71				
Firozpur	33,540	650,519	52	Total, British Territory	1,249,780	19,292,633	66

It will be seen that the number used per 1,000 of population varies enormously in different districts. This is partly due to the fact that, while in most Districts the more incompetent enumerators made a rough preliminary record and had it corrected by the supervisors before they copied it into their books, in some Districts this was written on schedules, and in others on paper ruled by hand. But there is no doubt whatever that in some Districts the indents were absurdly large and there was great waste. Many of the enumerators' books had 10 or 20 unused blank schedules at the end, and thousands of spare schedules were in some cases sent to the Divisional office after the Census to be used as blank paper. In many Districts, however, the Deputy Commissioners complain that the supply was insufficient. I think that in future a very liberal margin should be allowed for waste; far more liberal than was generally provided by Deputy Commissioners in their indents for this Census. I think that the supply should be nearly if not quite double that which would be needed if all enumerators wrote closely and neatly, and there was no waste. It is better that rough copies, if they are to be made, should be made on the printed schedule, as the enumerators thus grow accustomed to its use, and are saved both time and trouble, while the cost of the extra schedules is quite insignificant; and above all things the schedules should be distributed as early as possible, and every effort should be made to estimate and supply the deficiency, if any, at as early date as possible. At this Census District officers seemed to think that schedules grew at Lahore; and I received emergent indents for thousands of schedules up to within a few days of the actual Census.

947. Mr. Steedman has the following remarks on the subject :—

"Native subordinates send in the wildest indents. The only way to keep them in check is to require explanations of how the "forms previously supplied have been expended. A short time before the final enumeration I sent a circular round the tahsíls "asking the superintendents to send me in all surplus forms, and pointing out that according to the number supplied, the population "according to the business returns, and the number of enumerators, there ought to be a surplus. I believe this went a long way towards stopping unreasonable emergent indents at the last."

This is very true; but care must be taken not to make the officers feel that they may not use schedules for such purposes as rough copies and the like, or that there need be any hesitation about replacing a schedule which has been badly cut up by corrections. The relative competence of enumerators is almost as important a factor in deciding the number of schedules required as is the population to be enumerated. Where other characters than Persian are used, two copies, or an interlinear biglot copy occupying double space, must be made, and the practice of preparing a rough copy seems to have been very general. Mr. Tolbort of Ambála says: "At the commencement of this stage of the Census proceed-"ings I endeavoured to check the practice of preparing a rough record first and fairing it afterwards, but "the general consensus of opinion in favour of the said practice was so strong that I considered it unwise "to persist in opposing it. I do not think that it led to '*farsi*' procedure;" and most Deputy Commissioners agree with him, though Mr. Wilson of Sirsa prefers the original as being more trustworthy.

Part I.—Subsidiary Matters.

948. Expenditure.—The total expenditure on Census operations was as follows:—

Expenditure on the Census of 1881.

Enumeration of the People—	R
1. Pay and allowances of Deputy Superintendent	3,430
2. Office establishment	900
3. Telegrams	800
4. Paid agency and allowances	1,160
5. Paper for schedules and instructions	5,810
6. Printing schedules, typographed at Calcutta	750
7. Printing instructions, &c., lithographed at Lahore	3,650
8. Carriage of schedules from Calcutta by mail train	5,980
9. Boards for enumerators' books	7,800
10. Packing and distributing forms	2,520
11. District contingencies—Paper	4,090
12. Do. do. Carriage, &c.	2,180
13. Do. do. Postage and telegrams	1,200
14. Rewards to non-officials who had worked well	16,000
TOTAL ENUMERATION	56,370

Preparation of the Figures—	R
15. Paper for forms	4,850
16. Carriage of do.	1,230
17. Printing forms (lithography at Lahore)	6,540
18. Central office—Pay and allowances of Deputy Superintendent	10,040
19. Do. do. Office establishment	7, 50
20. Compiling office—Assistant in charge	5,200
21. Do. do. Establishment	7,320
22. Do. do. Contingencies	650
23. Divisional offices—Officers in charge	9,430
24. Do. do. Patwaris' substitutes	16,050
25. Do. do. Patwaris' allowances	18,870
26. Do. do. Clerks	1,830
27. Rent	730
28. Printing and Contingencies	3,300
TOTAL COMPILATION	93,090
TOTAL ENUMERATION	56,370
„ COMPILATION	93,090
29. Printing Report (estimated)	5,640
TOTAL EXPENDITURE	1,56,000

The total does not agree with that given on page 13, as after those figures were prepared an additional bonus equivalent to raising the monthly allowances of the patwáris employed in the Divisional offices from R2 to R4 was sanctioned, and the Deputy Commissioner of Bannu sent in a bill for expenditure nearly two years after date. From the latter cause, item 4, which is composed of R560 for supervision of the Teri Census in Kohát, R450 for paid enumerators in Bannu, and R150 travelling allowance paid to a European Superintendent of the School of Wards in Ambála, does not tally with the statement made in the body of the report about paid agency. Item 8 might have been reduced by three-quarters if we had begun printing early enough, and items 9, 10, 12, and 13 by nearly one-half had the binding boards for enumerators' books been made in the districts; while if items 20 and 21 had been a little larger, items 18 and 19 would have been much smaller. By having the printing done at private presses, items 7, 17, and 29 might be largely reduced on a future occasion.

949. Certificates and rewards.—After the operations were over, certificates to the following effect were distributed to those whose services deserved recognition: "This certificate is presented to_____by "order of the Hon'ble the Lieutenant-Governor of the Panjáb, in recognition of the good and willing service "he rendered in the Census operations of 1881, and as a mark of approbation." These certificates were handsomely printed in gold, red, and black on foolscap paper surmounted by the Royal Arms, and the three colours constituted three classes of certificates. Of these the first was granted to officials of the higher grades and natives of position who had done specially good work; the number issued was kept low so that a first-class certificate should be looked upon as a very special distinction. The second-class certificate was given to all persons who had distinguished themselves by zeal and activity; and the third class to all others whose services were considered worthy of recognition. The numbers distributed were as follows:—

1st Class	559
2nd Class	4,711
3rd Class	10,464
TOTAL	15,734

With these certificates small amounts from the grant of R16,000 made by the Government of India were distributed as rewards. None of this money was paid to any official or other person whose services were paid for by the State, or as a reward for work done in connection with the Census of a cantonment, or of a municipality liable for the cost of its own Census. The money was to be "expended in rewarding "selected men for especially good service, and not in general payment for work done." Thus the rewards were confined to non-officials who received second-class certificates, recipients of the first-class certificates being of too high standing to receive small money rewards, while those to whom third-class certificates were awarded were, *ex hypothesi*, not men selected for specially good service. The Government grant for rewards was in some districts supplemented by grants from Local and Municipal Funds.

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